MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCHIS101 - SCIENCE AND SOCIETY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Part A

Short Answer Type Questions

Answer any 5 out of 8 Questions Each Question carries 2 marks

1. Find the significance of Tycho Brahe in the history of science [U] / [CO1] – Set up an observatory on Hven Island in 1576 – observed planetary motion and distance between planets and stars – corrected ill notions on comets – mentor of Kepler

2. Explain the meaning of 'Cogito, ergo sum' [U] / [CO1] – Famous quote of Rene Descartes means ‘I think, therefore I am’ – found in his First Meditation titled 'What can be called into doubt’ – Descartes pointed to the employment of doubt towards the creation of knowledge.

3. Recall the contributions of Gottfried Leibniz to philosophy [U] / [CO1] – German philosopher – shared the invention of Calculus with Newton – active in the institutionalization of science in German region – the concept of monads as the fundamental building blocks of reality.

4. What is Bayt al-Hikmah? [K] / [CO2] – one of the largest libraries in the eastern region, located in Baghdad – Founded by Caliph Harun al-Rashid – Scholars like Al-Khindi and Al-Khwarizmi worked there.

5. What is Asvayurvedha? [U] / [CO2] – branch of ayurveda deals with the health and treatment of horses – fundamental part of Indian veterinary science - special medical care to horses because of its special role in travel and warfare.

6. What is the significance of 'Cinchona Plant'? [K] / [CO3] – Cinchona bark was used by the colonialists to prepare the anti-malarial drug quinine – crucial for European colonial powers to occupy and control tropical regions.

7. What is 'People's Science Movement'? [U] / [CO1, CO4] – Grassroot, community-based movements to popularize science, promote scientific thinking, counter anti scientific tendencies in society and use of science to address societal issues – use popular and innovative tools and mediums to spread scientific temper.

8. Trace the significant contributions of Albert Einstein to the knowledge and practice of science[U] / [CO4] – Special Relativity and General Relativity – Quantum theory of light – Bose-Einstein condensate -Highlighted the need of societal outlook of science in his article ‘Why Socialism’

Part B

Short Essay Type Questions

Answer any 4 out of 7 Questions Each Question carries 5 marks

9. What is normal science and which are the features of normal science? [K] / [CO1] – Normal science is an approach to the physical phenomena in a structural, natural and orderly manner – science in the 20th century can be treated as normal science – its main features can be locate in the nature of the function it provides which includes; core science (theoretical and experimental attempts on studying nature), applied science (making of technologies that useful to common man), professional consultancy (providing technical services) and managerial science (risk management for government and public)

10. Recall the role of William Harvey in revealing the functioning of blood circulation in body [K] / [CO1] – Harvey's theory disapproved the ancient Galenic model and laid foundation for modern physiology and medicine - Harvey identified the prime position of heart in blood circulation -. He discovered the flowing path of blood and role of vein and arteries in it in his treatise *De Mortu Cordis* – Harvey corrected the illogical understanding that blood is consumed and replenished by liver.

11. Analyze the influence of scientific method in the development of social science's methodology [An] / [CO2] – Social science greatly influenced by scientific methodology of observation and experimentation – Positivist logic has a direct scientific linage – Baconian and Cartesian methodologies provide new insight to the social scientists about theory building and enable to make methodical doubt – the craze for scientific objectivity became an enigma for many social scientists

12. Critically examine the terminology called 'Vedic mathematics' [An] / [CO2] – The term Vedic Mathematics is considered as a misnomer by many – the system is a collection of mathematical tricks and short keys compiled in the 20th century – there is no textual evidence to these methods in Vedas – An exception to this is only *Sulba* Sutra, deals with the conversion of rectangle to perfect square – India had a great tradition of mathematical practice and knowledge from the early days itself, but the tendency of ascribing nuances of mathematical tradition to a single category of Vedic mathematics in faux.

13. Explain how science has historically been influenced by social, cultural, and political factors [E] / [CO3] – The development of science in Europe and non-Western world was in tune with the social, political and cultural factors of the particular regions – gender, caste and Eurocentric stereotypes have an effect on scientific knowledge – Science in colonies often acted as an agent of colonialism and legitimized the racial outlook in colonial science – science represent the sophisticated mode of thinking of an age/society – Thus societal logic found an expression through science.

14. Explain the process of institutionalization of scientific practice in Europe [U] / [CO1] – Earliest associations for scientific transactions in Italy namely, Accademia dei Lincei (Rome) and Accademia del Cimento (Florence) - The Royal Society (England): Formed in 1660; monetary reliance on mercantile community - French Academy of Sciences: Founded in 1666 by Louis XIV and Colbert - The Parisian Académie Royale des Sciences was established in 1670 - Private Learned

Society, founded in 1774 in Bohemia – National Academy of Science in Russia in 1724 – in Belgium in 1772, Netherlands in 1808 – facilitated a space for scientists to gather together and discuss their knowledge.

15. Explain the reasons for the lack of representation of women and gender minorities in scientific research and profession [U] / [CO3, CO4] – Professionalization: As science became a paid profession, women were displaced and expected to be dependent - University Exclusion: Barred due to myths about biological inferiority (e.g., brain size) and reproductive risks - Closed Networks: Academies restricted membership to male "in-groups," excluding women - Masculine Ideology: Experimental science was framed as a masculine intervention into nature; women were deemed unfit - Attack on intellectual spaces: Female-led intellectual spaces (*salonnieres*) were vilified as chaotic and neglectful of domestic duties. Public vs. Private: Women were confined to the private/domestic sphere, rendering their "amateur" work unscientific.

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Part C

Essay Type Questions

Answer any 2 out of 4 Questions Each Question carries 10 marks

16. Recall the developments in natural science and astronomy in the pre-Newtonian Europe [U] / [CO1] - William Gilbert: discovered the magnetic force of the earth and opposition between poles - Roger Bacon: major contributions were in optics, and in the studies on light, sound, and matter - Jean Buridan, who developed the concept of impetus, the precursor of Newtonian inertia - Jacopo Dondi dell'Orologio, Richard of Wallingford, Jean Buridan, and Giovanni Dondi dell'Orologio: designed astronomical clocks, developed planetariums, and formulated experiment-supported explanations for natural phenomena – Nicolaus Copernicus: heliocentric argument – Bruno: advocate of heliocentrism - Tycho Brahe: instituted observatories and observed planetary motion – Kepler: three laws of planetary motion – Galileo : modified telescope – initiated observation and experimentation methodically – heliocentrism.

17. Analyze the development of scientific knowledge in Chinese society in the premodern times -[An] / [CO2] - Joseph Needham: His work ("Science and Civilization in China") debunked the myth that China contributed little to science, proving they were ahead of the West Pre-Renaissance - Paper & Printing: Paper was invented by Cai Lun (105 CE). Woodblock printing was followed by movable type in the 10th century. -Gunpowder: Discovered accidentally by Taoist alchemists (charcoal, Sulphur, saltpeter). Weaponized into bombs and barrel guns by 1000–1120 CE.- Porcelain: Developed during the Han dynasty; reached artistic peaks in the Ming dynasty and exported to Europe. - Magnetism: Used Earth's magnetic field for navigation (compass) and understood magnetic declination long before the West. - Maritime: Constructed the Grand Canal (1,776 km). Invented the sternpost rudder and paddle-wheel boats - Iron & Steel:

Developed double-acting piston bellows (continuous blast) and steel production via carbonation (6th century) - Mechanics: Invented the stirrup, wheelbarrow, efficient horse harness, and mechanisms to convert reciprocating motion to rotary motion - Mathematics: Used a decimal place-value system (bamboo rods) and sliding calipers, Liu Hui edited "The Nine Chapters on the Mathematical Art." Calculated Pi (π) to seven decimal places nine centuries before Europe.- Astronomy: Shi Shen (4th century BCE) created a star catalogue, Recorded the first solar eclipse (1361 BCE), sunspots (28 BCE), and supernova. Built complex Armillary Spheres and hydromechanical clocks (e.g., Kaifeng Tower, 1088 CE).

18. Critically examine various aspects associated with the Eurocentric discourses on modern science - [E] / [CO3] - The critique of Eurocentrism argues that the "Scientific Revolution" was not a unique European event but a synthesis of global knowledge that erased Islamic, Indian, and Chinese contributions - The "Universal" Trap: Western science is elevated as the sole objective truth, while non-Western systems are dismissed as "ethnoscience" or folklore -Historical Appropriation: Many "Western" breakthroughs have unacknowledged Eastern roots. For instance, the Copernican model utilized Persian mathematics (Tusi), and the "scientific method" was practiced by Ibn al-Haytham centuries before Galileo - "Standard" medical data is overwhelmingly based on European genetics, causing treatments to fail for non-white populations, while "helicopter research" exploits Global South resources without credit – Eurocentrism a colonial project still working to establish the European intellectual dominance of the world and establish non-European inferiority.

19. Analyse the epistemological changes happening in the world of science in the post normal times - [An] / [CO3, CO4] – Following the outbreak of Covid pandemic science forced to face the challenge of ‘complexity’ – space once occupied by old foundational principles like natural, linearity, certainty usurped by chaos/disorder, pure multiplicity, non-linearity, chance and uncertainty – notions like unity of science got collapsed – meta or big science of the last century dealt with pure, applied, regulatory or management issues – post normal times demand post normal approach to science – a new scientific approach which enables to understand and interact with the issue of complexity – big science managerial logic found stuck when it face issues like climate change, zoonotic outbreaks, natural calamities etc – there is a need to widened and make ready the tools and approaches of science to face the complexity of the universe and new challenges.